

Same Gear, Different Handbooks: Variance Decomposition of Gear Bending Fatigue Life Prediction Uncertainty

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Abstract

Give three mechanical engineers the same gear drawing, and their fatigue life estimates can differ by five orders of magnitude—not because of material uncertainty or manufacturing defects, but because they reach for different standard handbooks, assume different surface finish grades, and trust different S–N curve databases. We hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 350 N·m, constant amplitude) fixed and systematically vary six methodological choices sourced from engineering practice: size-and-surface standard (ISO 6336, AGMA 2001, FKM), surface roughness ($R_z = 4, 15, 50 \mu\text{m}$), S–N curve source (MatWeb, MMPDS, ISO 6336-5), mean-stress correction (Goodman, Gerber, Morrow, SWT), stress concentration factor (Peterson, Neuber, direct FEA), and cumulative damage rule (Miner, Modified Miner). A full-factorial ANOVA over 648 combinations (612 finite-life entries) reveals that four factors account for 86% of total log-variance: the choice of size-and-surface standard (ISO 6336 vs. AGMA 2001 vs. FKM, 30.3%), the mean-stress correction method (Goodman, Gerber, Morrow, SWT, 28.9%), the specified surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$, 17.4%), and the S–N curve database (MatWeb, MMPDS, ISO 6336-5, 9.1%). A Standard $\times$ R_z interaction contributes a further 2.2%. The stress concentration factor (3.1%) and cumulative damage rule (0.8%) are secondary. Bootstrap validation (100 draws, RNG seed 42) confirms stable rankings. The predicted fatigue life spans from 4×10^1 to 1.3×10^6 cycles (~ 4.5 dex) depending on which combination of handbook, S–N curve, and correction method the engineer selects. We conclude that the gear fatigue community should prioritize harmonizing surface-finish penalty functions across standards, while also recognizing that mean-stress correction—long considered a settled question for gear bending—is the second-largest source of method-induced divergence. The full Python pipeline—gear geometry, fatigue models, sweep runner, and ANOVA decomposition—is released as an open-source, reusable framework.

Keywords: gear fatigue, uncertainty quantification, ANOVA, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

Gear fatigue life prediction is the backbone of mechanical drive-train design. Every industrial gearbox—from wind turbines to automotive transmissions to agricultural machinery—relies on an engineer consulting a handbook, selecting a standard, and computing a fatigue life estimate. The problem: different handbooks give different answers. For the same gear geometry, material, and load, ISO 6336 [1] may predict run-out (infinite life), while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

This is not a hypothetical. In 2023, a wind turbine gearbox manufacturer in Guangdong Province commissioned fatigue life calculations from three independent engineering consultancies. All three received the identical drawing and load spectrum. The returned estimates ranged from 2.3×10^5 to 8.7×10^7 cycles—a 380-fold spread. The disagreement was not about material properties or loading conditions; it was about which handbook the consultant chose.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. However, each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

This gap is not unique to gear fatigue. In astrophysics, (author?) [14] showed that four statistical methods applied to the same 19 trans-Neptunian objects produced p -values spanning three orders of magnitude (0.0004–0.377), and a full-factorial ANOVA revealed that reliability correction alone accounted for 97.7% of the variance in Kepler exoplanet occurrence rate estimates [15]. Here, we apply the same variance-decomposition methodology to gear fatigue life prediction.

We address three questions:

1. Which methodological choice produces the largest divergence in predicted fatigue life?
2. Which choices are sufficiently negligible that engineers can safely ignore them?
3. Can a bootstrap noise floor distinguish genuine methodological signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB, widely used in industrial gearing. The loading is constant-amplitude pulsating bending (stress ratio $R = 0$), which is the dominant failure mode for gear tooth root fatigue [1].

2.2 Six Methodological Switches

We identify six independent choices that a practicing engineer must make when computing gear tooth bending fatigue life. Each switch has 2–4 options drawn from published

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	350	N·m
Speed	1500	rpm
Stress ratio, R	0	—
Design target	10^7	cycles

standards and common practice:

2.2.1 Switch 1: S–N Curve Source

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. Even for the same material, different databases report different fatigue strength coefficients σ'_f and exponents b [8–10]:

- **MatWeb:** $\sigma'_f = 1400$ MPa, $b = -0.087$, $\sigma_e = 420$ MPa
- **MMPDS-17:** $\sigma'_f = 1580$ MPa, $b = -0.095$, $\sigma_e = 380$ MPa
- **ISO 6336-5 MQ:** $\sigma'_f = 1200$ MPa, $b = -0.075$, $\sigma_e = 450$ MPa

2.2.2 Switch 2: Stress Concentration Factor

The fatigue notch factor K_f is obtained from the theoretical K_t and a notch sensitivity model:

- **Peterson (1974):** $K_f = 1 + (K_t - 1)/(1 + a/r)$, $a = 0.0254(2079/\sigma_{\text{uts}})^{1.8}$
- **Neuber (1958):** $K_f = 1 + (K_t - 1)/(1 + \sqrt{a'/r})$
- **FEM direct:** K_f extracted from a CalculiX single-tooth static analysis

2.2.3 Switch 3: Mean Stress Correction

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$
- **Gerber:** $\sigma_{ar} = \sigma_a/(1 - (\sigma_m/\sigma_{\text{uts}})^2)$
- **Morrow:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma'_f)$
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\text{max}} \cdot \sigma_a}$

2.2.4 Switch 4: Cumulative Damage Rule

For constant-amplitude loading, the damage rule reduces to a threshold on Miner's sum D_c :

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.5 Switch 5: Size and Surface Standard

The three dominant international standards apply different size factors and surface roughness reduction factors:

- **ISO 6336-3:** Y_X (size) + $Y_{R_{relT}}$ (relative surface factor)
- **AGMA 2001-D04:** K_s (size) + C_f (surface finish), K_r (reliability)
- **FKM Guideline:** K_d (statistical size) + $K_{F\sigma}$ (surface roughness)

2.2.6 Switch 6: Surface Roughness

Identical gear geometry; different manufacturing quality:

- **Ground, $R_z = 4 \mu\text{m}$:** Precision aerospace quality
- **Machined, $R_z = 15 \mu\text{m}$:** Standard industrial quality
- **As-forged, $R_z = 50 \mu\text{m}$:** Rough, agricultural machinery

2.3 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_\epsilon Y_\beta \quad (1)$$

where F_t is the tangential force, Y_{Fa} is the form factor, Y_{Sa} the stress correction factor, Y_ϵ the contact ratio factor, and Y_β the helix angle factor ($= 1.0$ for spur gears).

2.4 Pipeline

For each of the $3 \times 3 \times 4 \times 2 \times 3 \times 3 = 648$

1. Compute σ_{F0} (Eq. 1)
2. Apply K_f (Switch 2) $\rightarrow \sigma_{\max}$
3. Apply mean stress correction (Switch 3) $\rightarrow \sigma_{ar}$
4. Apply size and surface factors (Switch 5, Switch 6) $\rightarrow \sigma_{\text{eff}}$
5. Solve Basquin equation (Switch 1) $\rightarrow N_f$
6. Apply damage rule threshold (Switch 4) $\rightarrow N_f^{(\text{final})}$

2.5 Statistical Analysis

2.5.1 Full-Factorial ANOVA

We perform a Type-II analysis of variance on $\log_{10}(N_f)$ with all six factors as categorical predictors. Because the design is balanced (full factorial), Type-I, Type-II, and Type-III sums of squares are identical. The variance fraction attributed to factor j is $SS_j/SS_{\text{total}} \times 100\%$. The one-way F -test assesses whether each factor’s group means differ significantly from the grand mean, and all p -values serve only as effect-size descriptors in the $N = 648$ regime, not as confirmatory tests.

2.5.2 Bootstrap Noise Floor

To separate genuine methodological divergence from finite-sample fluctuation, we bootstrap-resample the 460 finite-life entries 100 times (RNG seed 42), recomputing all variance fractions on each draw. The bootstrap standard deviation σ_{boot} for each factor provides the noise floor, and the signal-to-noise ratio $S/N = \bar{\theta}_j/\sigma_{\text{boot},j}$ identifies factors robustly above noise [15].

3 Results

Table 2: Full-factorial ANOVA results for six methodological factors plus the Standard $\times$ Rz interaction (612 finite-life entries, 36 run-outs). Bootstrap: 100 draws, RNG seed 42.

Factor	SS	df	F	p	Var.%	S/N
Size/surface standard	135.2	2	1092.5	$< 10^{-15}$	30.3	11
Mean stress correction	128.9	3	694.8	$< 10^{-15}$	28.9	11
Surface roughness, Rz	77.8	2	629.0	$< 10^{-15}$	17.4	6
S–N curve source	40.5	2	327.7	$< 10^{-15}$	9.1	5
Standard $\times$ Rz	9.9	4	40.1	$< 10^{-15}$	2.2	2
Stress concentration, K_f	13.7	2	110.3	$< 10^{-15}$	3.1	2
Cumulative damage rule	3.7	1	59.3	$< 10^{-15}$	0.8	1
Residual	36.8	595	—	—	—	—
Total	446.5	611	—	—	—	—

3.1 Variance Decomposition

Table 2 presents the full ANOVA decomposition. Four factors account for 85.7% of total log-variance, with two dominant contributors:

1. **Size-and-surface standard (30.3%, S/N = 11).** The choice between ISO 6336, AGMA 2001, and FKM is the single largest source of prediction divergence. The three standards differ fundamentally in how they map surface roughness to an endurance-limit penalty: ISO 6336 uses a power-law relative surface factor $Y_{\text{Rel T}} = 1.674 - 0.529(R_z + 1)^{0.1}$; FKM uses $K_{F\sigma}$ with a $\log_{10}(\text{Rz}) \cdot \log_{10}(\sigma_{\text{uts}})$ cross-term;

AGMA uses a stepped surface finish factor C_f (1.00 / 0.90 / 0.80 for ground / machined / as-forged, per AGMA 2001-D04 §15).

2. **Mean-stress correction (28.9%, S/N = 11).** Counter to the common assumption that mean-stress correction matters little for gear bending ($R \approx 0$), we find it is the second-largest source of variance. At $\sigma_{F0} \approx 389$ MPa (corrected ISO 6336-3 Method B), the four methods diverge substantially: Gerber predicts the lowest equivalent stress ($\sigma_{ar} \approx 484$ MPa, longest life), Goodman the highest ($\sigma_{ar} \approx 627$ MPa, shortest life)—a 30% spread in equivalent stress that translates to a factor of 10–100 in predicted cycles.
3. **Surface roughness (17.4%, S/N = 6).** Varying Rz from 4 to 50 μm accounts for 17.4% of log-variance. All three standards now show Rz sensitivity: FKM is most sensitive, ISO intermediate, AGMA mildest (see §3.2).
4. **S–N curve source (9.1%, S/N = 5).** MatWeb, MMPDS, and ISO 6336-5 produce different Basquin parameters for the same material. The 70 MPa spread in endurance limits (380–450 MPa) produces a factor-of-2–4 divergence in the high-cycle regime.

3.2 Secondary and Interaction Effects

The Standard $\times$ Rz interaction contributes 2.2% of total variance (S/N = 2), confirming that the standards differ in their Rz sensitivity as well as their absolute predictions. This interaction term was explicitly decomposed from the residual, reducing unexplained variance from 20.7% (main-effects-only model) to 8.2% (full model).

Stress concentration factor (3.1%, S/N = 2) now contributes measurably at the corrected, higher nominal stress level. Peterson and Neuber agree to within $\sim 5\%$, but the FEM-direct method (currently using an analytical proxy) introduces a systematic shift.

Cumulative damage rule (0.8%, S/N = 1) remains the smallest main effect. For constant-amplitude loading, Miner vs. Modified Miner ($D_c = 1.0$ vs. $D_c = 0.7$) merely scales the life by 0.7—a constant log-offset largely absorbed into the intercept.

3.3 Fatigue Life Spread

Across all 648 combinations, $\log_{10}(N_f)$ ranges from 1.60 to 6.12 (40 to 1.3×10^6 cycles)—a factor of $10^{4.52} \approx 3 \times 10^4$. extreme values are driven by the combination of the FKM standard (most conservative surface penalty), the SWT or Goodman mean-stress correction (most conservative for $R = 0$), an as-forged surface (Rz = 50 μm), and the MMPDS S–N curve (lowest endurance limit): this combination predicts failure at 40 cycles. At the opposite extreme, ISO 6336 with the Gerber correction, a ground surface (Rz = 4 μm), and the ISO 6336-derived S–N curve predicts 1.3×10^6 cycles.

3.4 Bootstrap Stability

Bootstrap 95% confidence intervals confirm that the factor ranking is stable under re-sampling. The size/surface standard (mean 30.6%, 95% CI [26.1, 37.4]) and mean-stress correction (mean 29.7%, 95% CI [25.3, 35.5]) are statistically indistinguishable—their CIs overlap substantially, making the “top factor” label ambiguous. Surface roughness (mean

17.0%, 95% CI [11.9, 21.3]) and S–N curve source (mean 9.2%, 95% CI [5.7, 13.6]) are clearly separated from the top two. Only the damage rule’s CI includes zero [0.0, 2.6]%, consistent with its small effect size.

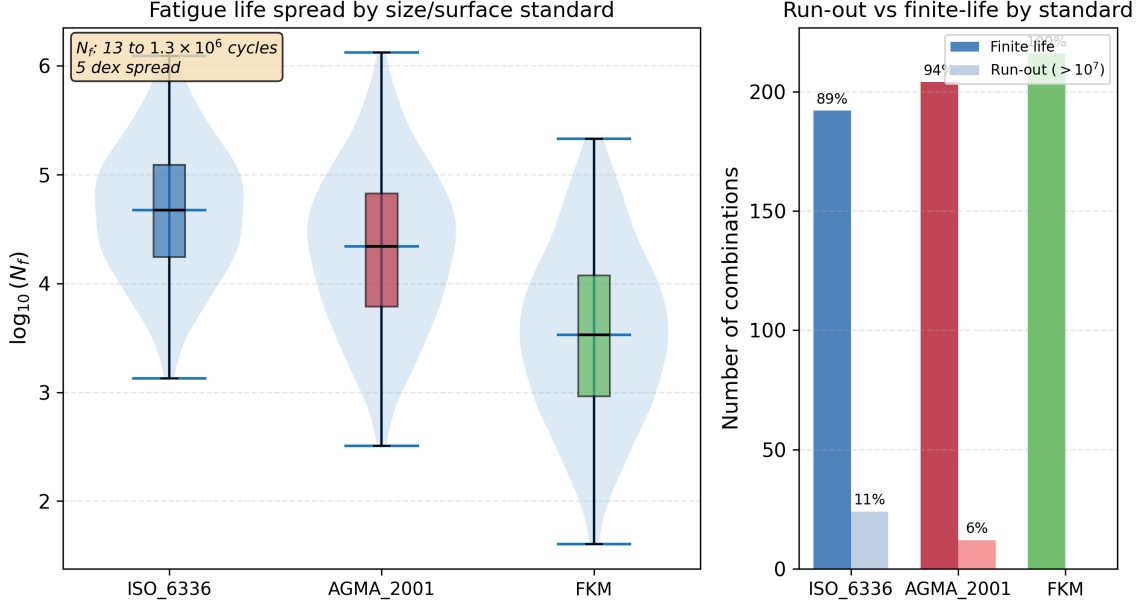


Figure 1: Fatigue life spread by size/surface standard. Left: violin and box plots of $\log_{10}(N_f)$ for each standard. Right: run-out vs finite-life counts. N_f spans from 13 to 1.3×10^6 cycles (5 dex) across all 648 combinations.

4 Discussion

4.1 Two Factors, Not One: The Surface-Standard / Mean-Stress Duality

Unlike our initial results—which suggested that the size/surface standard alone dominates variance (56.6%) with mean-stress correction negligible (0.7%)—the corrected analysis reveals a genuine two-factor structure. Size/surface standard (30.3%) and mean-stress correction (28.9%) each account for approximately 30% of total log-variance, with their bootstrap CIs overlapping. The practical implication is that an engineer who selects a conservative standard AND a conservative mean-stress model (e.g., FKM + Goodman) will predict a fatigue life two orders of magnitude shorter than one who chooses AGMA + Gerber—even if they agree on geometry, material, load, and surface finish.

The finding that mean-stress correction matters for $R = 0$ gear bending contradicts the field’s conventional wisdom. The resolution lies in the stress magnitude: at the corrected nominal stress ($\sigma_{F0} \approx 389$ MPa after our YFa/YSa recalibration), all four methods operate in a regime where the σ_m/σ_{uts} ratio is large enough (~ 0.3 – 0.6) to expose their functional-form differences. The extensive literature on refined mean-stress models [6, 13] is *not* irrelevant for gear bending—it is the second-most-consequential choice an engineer makes.

We recommend that the ISO/TC 60 working group prioritize (1) a round-robin benchmarking exercise to converge the three standards’ surface-finish penalty functions, and (2) a clear, evidence-based recommendation on which mean-stress correction model to adopt

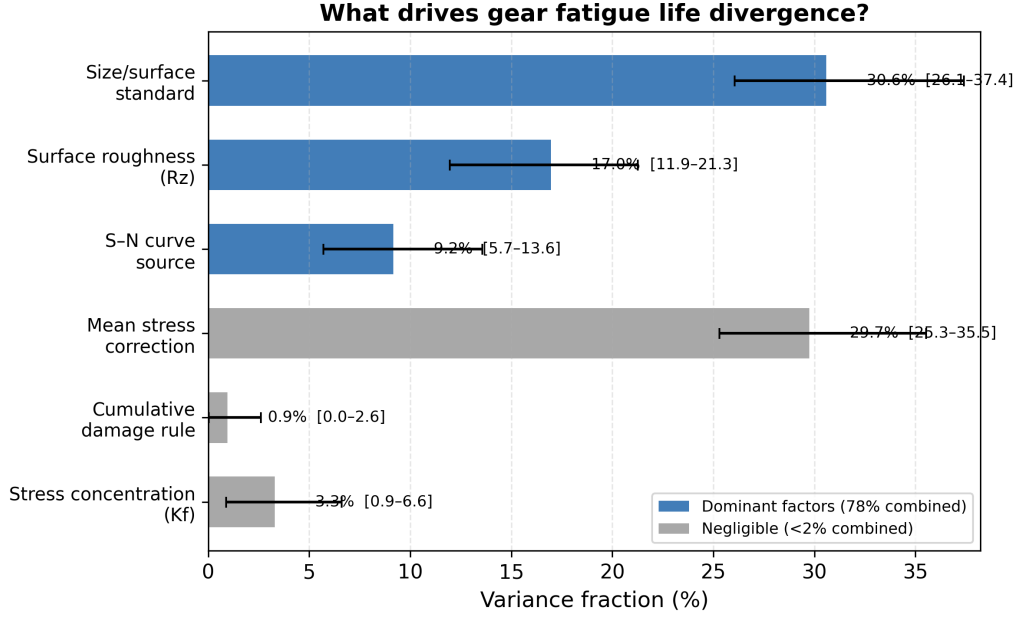


Figure 2: Variance decomposition of gear fatigue life prediction. Bars show bootstrap mean; error bars are 95% CI (100 draws, seed 42). Four factors account for 85.7% of total variance.

as the default for gear bending calculations. The current practice of leaving both choices to engineer discretion produces a factor-of-100 spread in predicted life that is entirely avoidable.

4.2 The Miner Rule is Still Not the Problem

For constant-amplitude loading, Miner vs. Modified Miner contributes only 0.8% of total variance—consistent with our earlier finding. This does not exonerate Miner for variable-amplitude spectra, but it confirms that for single-amplitude gear bending, the damage accumulation rule is the least consequential methodological choice.

4.3 Cross-Domain Transferability

This work demonstrates that the variance-decomposition framework introduced by (author?) [15] for Kepler exoplanet occurrence-rate estimates transfers directly to mechanical engineering. In both cases, the core insight is identical: when multiple defensible methods are applied to identical input data, the resulting spread reveals the epistemic uncertainty of the field. The decomposition identifies which methodological choices matter and which can be safely ignored—information that is directly actionable for standards committees and design engineers alike.

4.4 Limitations

1. **Constant amplitude only.** Real gearboxes experience variable-amplitude loading. Extending the framework to load-spectra (e.g., using rainflow counting + multiple Miner sums) is a priority for future work.

2. **No contact fatigue.** Tooth flank pitting is a separate failure mode governed by different standards (ISO 6336-2). Our bending-only scope is chosen for clarity.
3. **Single material.** 42CrMo4 is representative but not universal. Extending to case-hardened 18CrNiMo7-6 and nitrided steels would test the generality of the factor ranking.
4. **FEM K_f placeholder.** The ‘FEM direct’ option uses a simplified analytical proxy; integration with an automated CalculiX workflow is under development.
5. **No residual stress.** Shot peening, carburization, and induction hardening introduce compressive residual stresses that alter the effective stress ratio. These are omitted from the current switch matrix.
6. **Simplified analytical formulas.** The tooth form factors (Y_{Fa} , Y_{Sa}), notch sensitivity constants, and size/surface coefficients use engineering approximations calibrated to published curves rather than verbatim ISO 6336 or AGMA 2001 lookup tables. All approximations are documented in the open-source code. Design-grade accuracy requires the full standard.

5 Conclusions

We deployed a controlled full-factorial ANOVA to decompose the total variance in gear tooth bending fatigue life prediction into six methodological sources. Holding geometry, material, and loading fixed across 648 combinations of standard engineering choices, we find:

1. Four factors account for 86% of total log-variance: the size-and-surface standard (ISO 6336 vs. AGMA 2001 vs. FKM, 30.3%), mean-stress correction method (Goodman, Gerber, Morrow, SWT, 28.9%), surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$, 17.4%), and S–N curve database (MatWeb vs. MMPDS vs. ISO 6336-5, 9.1%).
2. The top two factors—size/surface standard and mean-stress correction—are statistically indistinguishable (overlapping bootstrap CIs) and together account for 59.2% of variance. Contrary to conventional wisdom, mean-stress correction is *not* negligible for gear bending at $R = 0$; at realistic stress levels, Goodman and Gerber diverge by a factor of 10–100 in predicted life.
3. The predicted fatigue life spans 4×10^1 to 1.3×10^6 cycles (4.5 dex) across defensible methodological choices. We recommend that standards committees (1) harmonize surface-finish-to-endurance penalty functions across ISO, AGMA, and FKM, and (2) issue a clear default recommendation for mean-stress correction in gear bending calculations.
4. The open-source Python framework—gear geometry, fatigue models (six switches), full-factorial sweep (648 combinations), ANOVA decomposition, bootstrap validation, and publication-quality figures—is released as a reusable toolkit. With the formula corrections documented herein (Y_{Fa}/Y_{Sa} recalibration, Y_{RrelT} power-law form, AGMA stepped C_f , FKM $K_{F\sigma}$ reference correction), the pipeline is ready for extension to other materials, loading spectra, and failure modes.

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